Intelligent Camera Surveillance System

Architecture Improvement Proposal

From a single PC to a distributed on-premises recording and AI analytics platform for 300–500 cameras

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Contents

Contents 2
1. Executive Summary 3
2. Current Situation and Root-Cause Analysis 3
2.1 Current configuration 3
2.2 Why a single PC cannot carry this load 4
2.3 Typical software anti-patterns that amplify the problem 5
3. The Big Picture: From the Current System to the Target Platform 5
3.1 Target architecture in one picture 5
3.2 Where each function of today's PC goes 6
3.3 What stays and what changes 7
3.4 Migration path 7
4. How Commercial VMS and Video-AI Systems Are Architected 8
4.1 Open-platform enterprise VMS (the reference architecture) 8
4.2 Enterprise security platforms with hard per-server limits 9
4.3 Lightweight / white-label VMS 9
4.4 AI network video recorders (appliance approach) 9
4.5 Edge analytics on the camera 9
4.6 NVIDIA Metropolis / DeepStream (the AI-analytics reference stack) 10
4.7 Common architectural pattern 10
5. Target Architecture — Detailed Design 11
5.1 Tier-by-tier specification 11
5.2 Capacity sizing 12
5.3 Reference hardware bill of materials (500 cameras) 12
5.4 Software architecture of the AI pipeline 13
5.5 Storage and network design notes 14
6. Dedicated Equipment for Camera Surveillance and Video AI 14
6.1 GPU and NPU accelerators for server-side analytics 14
6.2 Purpose-built recording servers and appliances 15
6.3 Edge-AI cameras (analytics on the camera) 15
6.4 Commercial software platforms worth evaluating 16
7. Implementation Roadmap, Budget, and Risks 16
7.1 Indicative budget envelope (500 cameras, hardware only) 17
7.2 Risks and mitigations 17
8. Conclusion 17

1. Executive Summary

“Company A” currently operates an intelligent camera surveillance system in which several hundred IP cameras (300–500 units, 1080p to 4K) are connected to a single PC that performs video ingest, recording, and inference with multiple AI models. The machine reaches 100 % CPU, GPU, and memory utilisation and becomes unresponsive, and upgrading the specification of that one PC has not solved the problem.

This is not a defect of the PC; it is a structural ceiling. No commercial video management system (VMS) or AI video analytics product runs several hundred cameras with multi-model AI on one machine. The leading open-platform VMS vendors state that a single server is suitable for roughly 50–100 cameras and that the cost/performance sweet spot is 200–300 cameras per dedicated recording server, with AI analytics moved to separate GPU servers. Enterprise VMS products publish hard per-server limits of about 100 cameras / 200 Mbps per recording node. NVIDIA’s own DeepStream benchmarks show that even a data-centre GPU such as the L4 sustains about 80 concurrent 1080p30 streams with a single lightweight detection model, so 400 cameras with several models on one PC is physically impossible.

We therefore recommend a distributed on-premises architecture with clearly separated tiers: a management/database tier, a recording tier of two to three recording servers, an AI analytics tier of two to four GPU servers running a batched inference pipeline, a two-tier storage system (SSD live recording plus HDD archive), and a dedicated surveillance network with 10 GbE server uplinks. Where possible, first-pass detection should be pushed to AI-capable cameras (edge analytics) so that servers only process metadata and events rather than every raw frame. Section 3 gives the complete picture of how the current system maps onto this target and how the migration proceeds without service interruption.

Key recommendations at a glance

Stop scaling up, start scaling out. Split ingest/recording, AI inference, management/database, and storage onto separate machines, each sized to a known, published capacity limit.

Move AI to a GPU analytics cluster. Run inference with NVIDIA DeepStream / TensorRT on data-centre GPUs (NVIDIA L4 class) using batched, reduced-frame-rate, sub-stream processing rather than decoding every full-resolution frame.

Use dedicated equipment. Purpose-built storage-dense recording servers, GPU/NPU accelerator cards, AI-enabled network video recorders (NVRs), and edge-AI cameras exist precisely for this workload.

Phase the migration. Immediate stabilisation within 2–4 weeks (frame-rate and sub-stream changes), server split within 1–3 months, GPU cluster and edge AI within 3–12 months, while the existing PC keeps running a reduced camera set.

2. Current Situation and Root-Cause Analysis

2.1 Current configuration

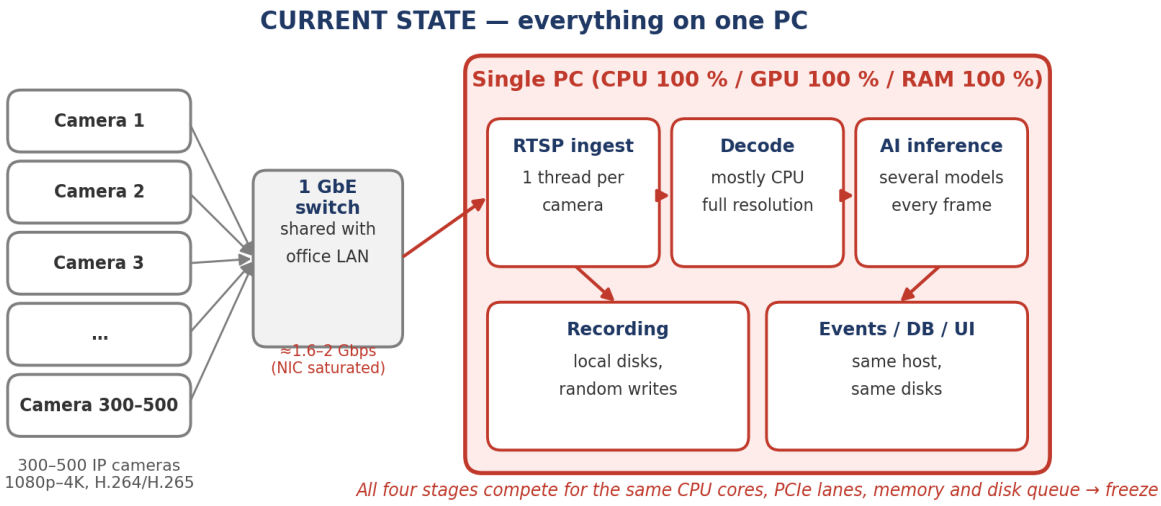


Figure 1 – Current architecture: every processing stage for every camera runs on one machine.

Item	Current state
Cameras	300–500 IP cameras, 1080p to 4K, H.264/H.265
Processing node	One PC handling ingest, decoding, recording, and inference
AI workload	Multiple AI models (e.g., person/vehicle detection, tracking, recognition) applied to camera streams
Symptom	CPU, GPU, and memory at 100 %; system freezes / stops responding
Attempted fix	Hardware upgrade of the single PC — no lasting improvement

2.2 Why a single PC cannot carry this load

Each camera stream must pass through four resource-hungry stages: network ingest, video decoding, AI inference, and disk writing. The table below estimates the aggregate load for 400 cameras at 1080p/30 fps with a typical 4 Mbps H.264/H.265 bitrate. The figures are conservative and scale linearly; 4K cameras multiply decode and inference cost by roughly four.

Resource	Per camera (1080p30, 4 Mbps)	400 cameras	What one PC can realistically do
Network ingest	4 Mbps	≈ 1.6 Gbps sustained	1 GbE NIC saturates at ≈ 0.8–0.9 Gbps usable; needs 10 GbE
Video decode	30 frames/s	12,000 frames/s	One NVDEC-class GPU decoder block handles ≈ 700–900 fps of 1080p H.264 (NVIDIA Video Codec SDK figures), i.e. ≈ 25–30 full-rate streams per decoder; CPU software decode is far lower
AI inference	30 inferences/s per model	12,000 × number of models per second	NVIDIA L4 ≈ 81 streams at 1080p30 with ONE ResNet-18 detector (DeepStream benchmark); several models cut this to 20–40 streams

Resource	Per camera (1080p30, 4 Mbps)	400 cameras	What one PC can realistically do
Disk write	0.5 MB/s	≈ 200 MB/s continuous, ≈ 17 TB/day	Feasible only with a RAID array of enterprise disks; consumer SSD/HDD in a PC will throttle and fragment
Memory	≈ 100–300 MB per decoded pipeline	40–120 GB	Exceeds most workstation RAM; swapping leads to the observed freeze

Because the four stages compete for the same CPU cores, PCIe lanes, memory bandwidth, and disk queues, the machine degrades non-linearly: once any one resource saturates, decode queues back up, RTSP buffers overflow, the inference batch waits, and the OS eventually stops scheduling the application. A faster CPU or a bigger GPU only moves the ceiling by a few dozen cameras; it does not remove it.

2.3 Typical software anti-patterns that amplify the problem

Beyond raw capacity, most in-house systems that freeze at scale share several design patterns. Each of these should be checked in the current code base:

- **Decoding the main (full-resolution) stream for analytics.** Most detection models run on 640×640 or smaller inputs; decoding 1080p/4K just to downscale wastes 4–16× decode capacity. Industry practice is to analyse the camera's low-resolution sub-stream.
- **Running inference on every frame.** Security analytics rarely need more than 5–10 fps; commercial systems skip frames or infer on motion events only.
- **One process or thread per camera with its own model copy.** Loads the GPU model hundreds of times and thrashes VRAM; a batched pipeline (DeepStream, Triton) loads each model once and infers many streams per batch.
- **CPU software decoding (FFmpeg/OpenCV default).** Hardware decoders (NVDEC on GPUs, the media engine integrated in server CPUs) must be enabled explicitly.
- **Recording and analytics on the same disk and the same host.** Random-write load from hundreds of recording files starves the OS and the database.
- **No back-pressure or watchdog.** When a stream stalls, buffers grow until memory is exhausted; commercial VMS drop frames gracefully and restart individual pipelines.

3. The Big Picture: From the Current System to the Target Platform

This section is the overall map of the change. It shows the target architecture as one picture, explains how each function that exists today on the single PC is relocated, and shows the order in which the pieces are introduced. Sections 4–6 then provide the evidence, the detailed sizing, and the equipment options.

3.1 Target architecture in one picture

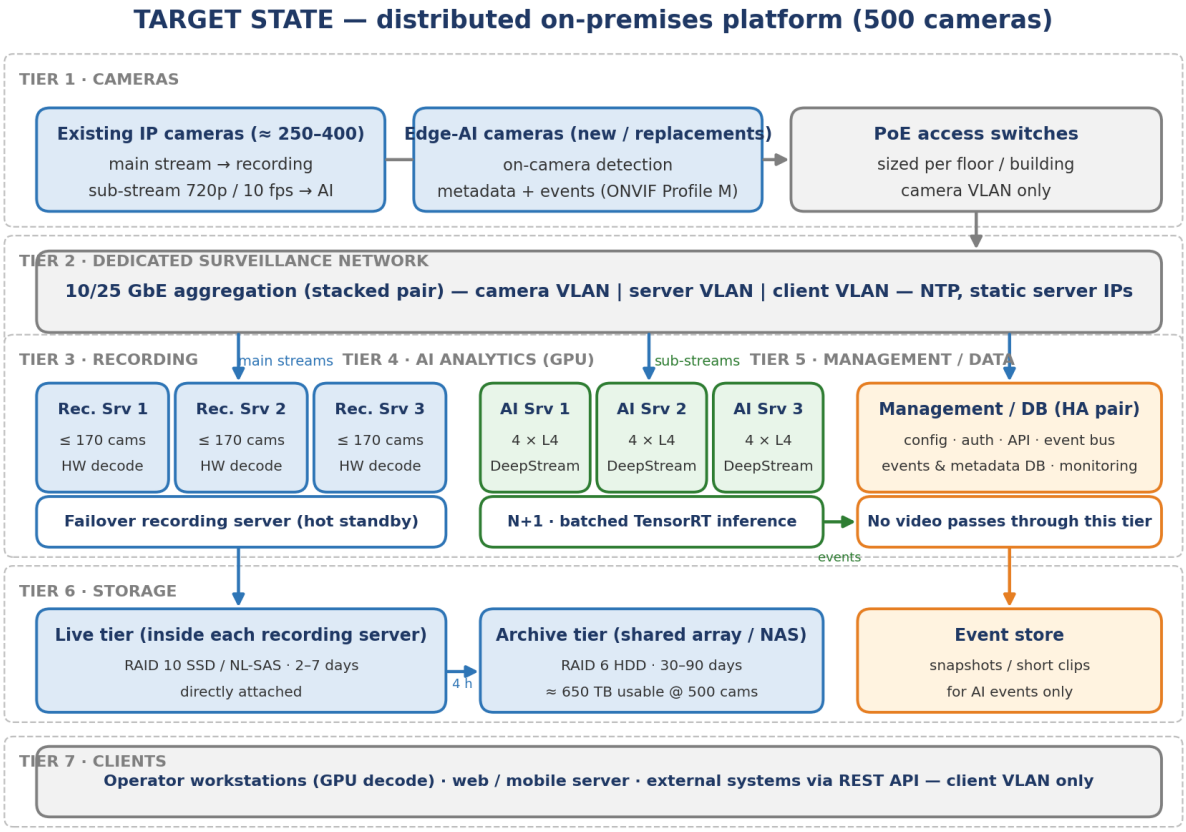


Figure 2 – Target architecture: seven tiers, each on its own hardware and each sized to a published capacity limit.

Video flows top-down: cameras send a full-resolution main stream to the recording tier and a small sub-stream (or, for edge-AI cameras, only metadata) to the AI tier. Only events and metadata flow to the management tier; no video ever passes through it. Recordings move from the fast live tier to the archive tier every few hours in sequential bulk transfers. Operators and external systems connect on a separate client network and never touch the camera network.

3.2 Where each function of today's PC goes

Function on the current PC	Target location	Why it moves	Sizing rule
RTSP ingest + recording of all cameras	Recording tier: 2–3 recording servers + 1 hot-standby failover	Recording is disk- and network-bound; it must not share a host with GPU inference	≤ 150–200 cameras per server; ≈ 0.8 Gbps per server
Decoding of full-resolution frames for AI	Eliminated: AI uses camera sub-streams (720p / 5–10 fps); recording servers decode only for motion detection using hardware decoders	Removes 4–16× of decode work	Sub-stream ≤ 1280×720, 5–10 fps
AI inference (all models, all frames)	AI analytics tier: 2–3 GPU servers, each with 4 × NVIDIA L4, DeepStream + TensorRT, batched	Inference scales with GPUs; batching loads each model once for hundreds of streams	≈ 40–50 streams per L4 with 2 models; N+1 servers

Function on the current PC	Target location	Why it moves	Sizing rule
Events, database, user interface, API	Management / data tier (HA pair): configuration, authentication, REST API, event bus, metadata DB, monitoring	Control plane must stay responsive even when video load peaks	One HA pair per site; DB on its own host above 300 cameras
Local disks holding recordings and DB	Storage tier: RAID 10 SSD/NL-SAS live tier inside each recording server; shared RAID 6 archive array	Sequential archive writes on cheap disks; random live writes on fast disks	Live 2–7 days, archive 30–90 days
1 GbE office switch	Dedicated surveillance network: PoE access switches, 10/25 GbE aggregation, separate camera / server / client VLANs	≈ 2 Gbps ingest plus playback and archive traffic exceed 1 GbE	10 GbE per server, dual NICs
Operator viewing on the same PC	Client tier: operator workstations with GPU decode, web/mobile server	Viewing load must not compete with recording	Multicast or dedicated streaming server for many viewers
(nothing today)	Edge-AI cameras for new installations and replacements	First-pass detection on the camera removes decode + inference from servers	Reduces GPU need to ≈ 4–6 × L4 at 500 cameras once half the fleet is edge-AI

3.3 What stays and what changes

Kept as-is	Changed	New
Existing cameras and cabling Existing AI models (re-exported to TensorRT/ONNX) Existing event schema and user-facing features The current PC (re-used as a client workstation or test node)	Analytics input: main stream → sub-stream Inference: per-camera loops → batched GPU pipeline Recording: local disks → dedicated recording servers with RAID Network: shared 1 GbE → dedicated VLANs with 10 GbE uplinks	Recording servers + failover GPU analytics servers (NVIDIA L4) Management/DB HA pair with event bus and monitoring Archive storage array Edge-AI cameras (phased)

3.4 Migration path

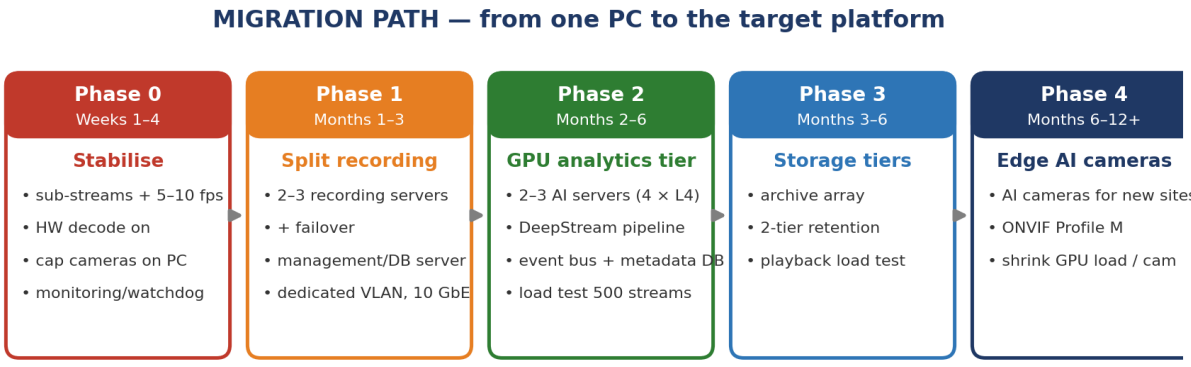


Figure 3 – Five-phase migration; capacity is added by adding nodes, never by replacing a single machine.

The order of the phases is chosen so that risk falls fastest first. Phase 0 changes only software settings and stops the freezes within weeks. Phase 1 makes recording independent of AI load, which is the most important reliability gain. Phase 2 restores full AI coverage on proper hardware. Phases 3 and 4 optimise cost and prepare for growth. Throughout, the existing PC continues to serve a reduced set of cameras until its cameras have been migrated in batches of roughly 50.

4. How Commercial VMS and Video-AI Systems Are Architected

This section summarises, with published figures, how established products structure deployments of several hundred cameras. Vendor and product names are omitted deliberately; the figures come from publicly available sizing guides, white papers, and benchmarks of the leading products in each category. The common thread is separation of roles across multiple servers, each with a documented capacity limit, plus hardware-accelerated decoding and dedicated GPU or NPU analytics nodes.

4.1 Open-platform enterprise VMS (the reference architecture)

The most widely deployed open-platform VMS products, which most system integrators follow as a reference, are built from deliberately separable components:

- **Management server** (configuration, authentication, relational database) — separate host for any system above about 100 cameras; the database on its own host above roughly 300 cameras.
- **Recording servers** (ingest, motion detection, recording to a media database) — published storage guidance places the sweet spot between performance and cost at 200–300 cameras per recording server; higher densities cause disk fragmentation and performance decay. Validated reference builds on storage-dense 2U servers record 120 cameras at 1080p30 / 4 Mbps at only 4–7 % CPU with ≈ 160 MB/s disk throughput — disk capacity and I/O, not CPU, is the limit.
- **Failover recording servers** — hot-standby servers that automatically take over a failed recording server's cameras.

- **Hardware acceleration** — CPU-integrated media engines and NVIDIA data-centre GPUs with NVDEC are recommended on recording servers for motion-detection decoding; consumer gaming GPUs are explicitly not recommended for sustained server use.
- **AI analytics gateway** — AI does not run inside the recording server. An API gateway forwards video to separate GPU-accelerated, container-based analytics servers (NVIDIA Metropolis / DeepStream stack) and returns events and metadata to the VMS.
- **Storage tiers** — live recording database on RAID 10 enterprise SSD, directly attached; archive on RAID 5/6 HDD or NAS/SAN, archived every ~4 hours; large file-system allocation units and RAID stripe sizes ≥ 512 kB.

One published multi-GPU benchmark recorded 1,000 Full-HD H.264/H.265 cameras on a single surveillance-grade server, but that is a **recording-only** result using several GPUs purely for hardware decode and motion detection — not multi-model AI inference.

4.2 Enterprise security platforms with hard per-server limits

Enterprise unified security platforms publish explicit ceilings. A representative current specification rates a combined directory + recording role at 50 cameras / 50 Mbps on the minimum server profile and 100 cameras / 200 Mbps on the recommended profile; a dedicated stream-redirection role handles up to ≈ 475 cameras / 475 Mbps. The recommended server profile (10-core server CPU, 16 GB RAM, GbE) is rated for up to 300 Mbps of video. Enabling stream encryption reduces recording capacity by 30 % or more. A 400-camera site is therefore expected to use four or more recording servers plus a directory server.

4.3 Lightweight / white-label VMS

Widely white-labelled media-server VMS products limit each server to 256 resources (cameras, encoders, I/O devices) on 64-bit builds and advise leaving headroom for failover. Each camera can spawn up to four threads (high/low stream read and container conversion), so a 256-camera server may run ≈ 1,000 active threads — an illustration of why hundreds of cameras on a desktop OS saturate the scheduler. The same vendor has validated NPU-based analytics servers (16 low-power AI accelerator chips in a 2U chassis) processing more than 100 AI channels at a quoted fraction of the cost of a GPU solution.

4.4 AI network video recorders (appliance approach)

Major camera manufacturers sell AI NVR appliances with embedded deep-learning engines. A representative 32-channel unit accepts 320 Mbps of incoming video, decodes 32 × 1080p30, and provides face-picture comparison on up to 32 channels, facial detection on 16 × 2 MP channels, perimeter protection on 24 × 2 MP channels, and video structuralisation on 12 × 2 MP channels, with 8 disk bays at ≤ 50 W. The design lesson is explicit: a professional vendor allocates roughly one AI recorder per 32 cameras, with fixed per-engine analytics channel budgets — not one box per 400 cameras.

4.5 Edge analytics on the camera

The leading camera manufacturers have pushed AI into the camera itself. Current camera system-on-chip generations include a deep-learning processing unit (DLPU) running object detection and

classification on-camera; the newest generation (first products shipped in 2025) triples analytics performance and adds AV1 encoding with content-adaptive bitrate reduction of ≈ 20 %. In an edge-analytics architecture, cameras detect and classify objects and stream metadata (ONVIF Profile M); the server only records and correlates events, removing the decode-plus-inference cost for the majority of cameras.

4.6 NVIDIA Metropolis / DeepStream (the AI-analytics reference stack)

NVIDIA's DeepStream SDK is the de-facto pipeline for multi-stream inference and is what VMS vendors and most analytics software vendors build on. Its published benchmark (DeepStream 7.0, TrafficCamNet ResNet-18 detector with tracker, 1080p H.265 at 30 fps) gives a realistic per-device capacity:

Platform	1080p30 streams (1 detector + tracker)	Notes
NVIDIA Jetson Orin Nano	13	Edge module, 7–15 W
NVIDIA Jetson Orin NX	16	Edge module
NVIDIA Jetson AGX Orin	37	Edge server, 15–60 W
NVIDIA A2	31	Low-profile 40–60 W PCIe
NVIDIA T4	45	Previous-generation 70 W PCIe
NVIDIA L4	81	Single-slot 72 W PCIe, 4 × NVDEC, 24 GB
NVIDIA A10	79	150 W PCIe
NVIDIA A30	150	165 W PCIe
NVIDIA L40	166	300 W PCIe
NVIDIA A100 PCIe	180	300 W PCIe
NVIDIA H100	229	350 W+

NVIDIA engineers also note that NVDEC decoder throughput is only one limit; scaling, colour conversion, and model compute share the GPU's streaming multiprocessors, so real pipelines with several or larger models achieve well below these numbers. Practical planning figures used by integrators are 30–50 cameras per L4-class GPU for two or three concurrent models at 5–10 fps analysis rate.

4.7 Common architectural pattern

Across all of these product categories the same layered design appears, and it is the pattern this proposal adopts:

Tier	Role	Typical scale unit	Product category
Edge	Encode, optional on-camera analytics, sub-streams	Per camera	Edge-AI cameras with DLPU
Recording / ingest	RTSP ingest, motion detection, write to media DB	100–300 cameras per server	VMS recording server / archiver

Tier	Role	Typical scale unit	Product category
AI analytics	Batched GPU/NPU inference, tracking, events	30–80 streams per GPU, 2–4 GPUs per server	NVIDIA DeepStream/Metropolis servers, NPU servers
Management / data	Configuration, users, events DB, API, message bus	One (HA pair) per site	VMS management server + database
Storage	Live SSD tier + HDD/NAS archive tier	Sized by retention	RAID 10 live + RAID 6 archive arrays
Clients	Operator viewing, mobile	Separate from servers	Desktop client, web/mobile server

5. Target Architecture — Detailed Design

The target design keeps everything on-premises, reuses existing cameras and network cabling wherever possible, and replaces the single PC with a small cluster of purpose-specific servers. It is dimensioned for 500 cameras with headroom, and can be commissioned in stages while the existing PC continues to operate on a reduced camera set.

5.1 Tier-by-tier specification

Tier	Components	Function
1. Cameras	Existing 300–500 IP cameras; new cameras specified with on-board AI (DLPU class)	Deliver main stream (recording) and sub-stream (640×360 to 1280×720, 5–10 fps) for analytics; AI cameras publish object metadata/events via ONVIF Profile M
2. Network	Dedicated surveillance VLANs, PoE access switches, 10/25 GbE aggregation, dual 10 GbE server NICs (camera network vs. client/storage network)	Isolate ≈ 2 Gbps camera traffic from the office LAN
3. Recording tier	2–3 storage-dense 2U recording servers, each ≤ 200 cameras, hardware decoder (CPU media engine or NVIDIA L4) for motion detection; 1 failover recording server	Ingest, record, index, serve playback; never runs AI models
4. AI analytics tier	2–4 GPU servers (Linux, containers), each 2–4 × NVIDIA L4 (NPU accelerator cards as an alternative), DeepStream + TensorRT/Triton	Pull sub-streams, batched inference, tracking, event generation; scales horizontally by adding GPUs/servers
5. Management & data	1 management server (HA pair recommended): configuration, auth, REST API, event bus (MQTT/Kafka), relational DB (e.g., PostgreSQL) for events & metadata, metrics/monitoring stack	Single control plane; no video passes through it
6. Storage	Live: RAID 10 enterprise SSD/NL-SAS inside each recording server (2–7 days). Archive: shared RAID 6 array or NAS for 30–90 days	Two-tier, sequential archive moves; snapshots/clips for AI events kept separately

Tier	Components	Function
7. Clients	Operator workstations with GPU decode, web/mobile server	Viewing load kept off servers via multicast or a dedicated streaming server

5.2 Capacity sizing

The following sizing uses published limits with a 25–30 % safety margin. It assumes analytics on sub-streams at 10 fps with two concurrent models (e.g., a person/vehicle detector plus a secondary classifier or re-identification model). If four or more models must run on every camera, add one GPU server per additional two models.

Parameter	300 cameras	500 cameras	Basis
Recording servers	2 (≈150 each) + 1 failover	3 (≈167 each) + 1 failover	200–300/server sweet spot with margin; enterprise limit 100–200/server
Recording ingest bandwidth	≈ 1.2 Gbps	≈ 2.0 Gbps	4 Mbps/camera average; 4K cameras at 8–12 Mbps
Live storage (7 days, SSD/NL-SAS RAID 10)	≈ 90 TB	≈ 150 TB	0.5 MB/s/camera × 86,400 s × 7
Archive storage (30 days)	≈ 390 TB usable	≈ 650 TB usable	Same rate × 30; H.265 and motion-only recording can halve this
GPU inference (2 models, 10 fps sub-stream)	6–8 × L4	10–12 × L4	≈ 40–50 effective streams/L4 (DeepStream 81-stream benchmark de-rated for 2 models and overhead)
AI servers	2 × (4 × L4)	3 × (4 × L4)	N+1 recommended so one server can be lost
Management/DB server	1 (HA pair optional)	1 HA pair	DB separate above 300 cameras (VMS guidance)
Network core	10 GbE aggregation, 10 GbE server NICs	25 GbE aggregation, dual 10 GbE server NICs	Ingest + playback + archive moves

Effect of edge-AI cameras on sizing

If 50 % of cameras are replaced over time with on-board-AI models, those cameras deliver detection metadata directly and the GPU tier only handles second-stage analytics (re-identification, counting, face/plate matching) on events. In that scenario the 500-camera GPU requirement falls to roughly 4–6 × L4, and the server count stays at two AI servers.

5.3 Reference hardware bill of materials (500 cameras)

Qty	Item	Indicative specification	Purpose
3 + 1	Recording server	2U storage-dense rack server, 2 × server-class CPUs (16–24 cores each), 128 GB RAM, 24 × 16–20 TB NL-SAS RAID 6 + 4 × SSD RAID 10 (live DB), dual 10 GbE, hardware RAID controller with cache; optional NVIDIA L4 or CPU media engine for hardware motion decode	Ingest & recording ≤ 170 cameras each; one hot failover
3	AI analytics server	2U GPU-capable rack server, 2 × server-class CPUs, 256 GB RAM, 4 × NVIDIA L4 24 GB, 2 × NVMe, dual 25 GbE; Linux, containers, DeepStream 7.x, Triton	Batched inference for 500 sub-streams, N+1
2	Management / DB server	1U server, 16–32 cores, 64–128 GB RAM, NVMe RAID 1; PostgreSQL or equivalent; MQTT/Kafka; metrics and dashboards	Control plane, events, API (HA pair)
1	Archive storage	Enterprise disk array or scale-out NAS, ≈ 84 × 20 TB, ~650 TB usable, 12 Gb SAS or 25 GbE iSCSI	30-day archive
1	Network core	2 × 25/10 GbE aggregation switches (stacked), PoE+ access switches sized for camera count	Dedicated surveillance VLANs
1	UPS + PDU	Rack UPS sized for ≈ 6–8 kW with 15–30 min runtime	Clean shutdown, no corrupted recordings

5.4 Software architecture of the AI pipeline

The analytics tier should be re-implemented (or wrapped) as a batched streaming pipeline rather than per-camera loops. The following practices are what commercial analytics vendors do and account for most of the difference between a system that freezes at 100 cameras and one that runs 500:

1. **Hardware decode only.** Use NVDEC via DeepStream (nvv4l2decoder) or FFmpeg with h264_cuvid/hevc_cuvid, or the CPU-integrated media engine (VA-API). Never use CPU software decode for more than a handful of streams.
2. **Analyse the sub-stream.** Configure each camera's second stream at 640×360–1280×720, 5–10 fps, and point the AI pipeline at it. Decode cost drops 4–16×; detection accuracy for people/vehicles is unchanged for most models.
3. **Batch inference with TensorRT.** Load each model once per GPU, infer on batches of 16–64 frames from many cameras (DeepStream nvinfer / Triton Inference Server). Convert models to TensorRT FP16 or INT8.
4. **Skip frames and cascade models.** Run the primary detector at 5–10 fps; run secondary models (classification, attributes, re-ID, face/plate) only on detected objects or on motion events, not on every frame.
5. **Zero-copy on GPU.** Keep frames in GPU memory from decode through pre-processing, inference, and tracking; copy only metadata to the CPU.
6. **Event-driven outputs.** Publish detections as JSON metadata to MQTT/Kafka; store snapshots and short clips, not continuous annotated video.

7. **Back-pressure, watchdogs, and isolation.** Bounded queues per stream, automatic pipeline restart per camera on stall, containerised workers so one faulty camera cannot take down the node.
8. **Observability.** Export per-stream fps, decode latency, GPU utilisation (NVIDIA DCGM exporter), disk queue depth, and dropped-frame counters to a metrics system; alert before saturation.
9. **Horizontal scaling.** Assign cameras to GPU workers via a scheduler (Kubernetes or a simple assignment table); add a GPU or a server to add capacity, with no code changes.

5.5 Storage and network design notes

- Live recording database on directly attached RAID 10 (SSD or fast NL-SAS), file system with large allocation units; archive to RAID 6 / NAS every few hours. Never record directly to a network share or USB disk.
- Retention policy by camera class (e.g., 30 days for entrances, 14 days for general areas) can reduce archive capacity by 30–50 %.
- Motion-based or event-based recording with a pre/post buffer, where acceptable, typically halves storage versus continuous recording.
- Two physical networks per server: camera-facing and client/storage-facing. Avoid Layer-3 routing inside the camera LAN; use static IPs for servers and NTP for all devices.
- H.265 (or AV1 on the newest edge-AI cameras) reduces bitrate 20–50 % versus H.264 at equal quality, which lowers every downstream cost.

6. Dedicated Equipment for Camera Surveillance and Video AI

The following equipment categories are specifically designed for multi-camera recording and video analytics. Indicative prices are approximate list-price ranges as of 2025–2026 and must be confirmed with local distributors; they are included only to support budgeting.

6.1 GPU and NPU accelerators for server-side analytics

Product / category	Key specifications	Streams (1080p30, 1 detector)	Indicative price	Fit for “Company A”
NVIDIA L4	Ada architecture, 24 GB, 72 W, single-slot low profile, 4 × NVDEC / 2 × NVENC, AV1/H.265 decode	≈ 81 (DeepStream benchmark)	USD 2,500–3,500	Primary recommendation: best streams-per-watt, fits 1U/2U servers, 4–8 per server
NVIDIA A2	Ampere, 16 GB, 40–60 W, low profile	≈ 31	USD 1,200–1,800	Small sites or recording-server motion decode
NVIDIA RTX 6000 Ada / L40S	48 GB, 300–350 W, 3 × NVDEC	≈ 166 (L40)	USD 7,000–10,000	When large models (vision-language, transformer re-ID) are required

Product / category	Key specifications	Streams (1080p30, 1 detector)	Indicative price	Fit for "Company A"
NVIDIA Jetson AGX Orin (industrial box PC)	64 GB, 15–60 W, ARM, fanless options	≈ 37	USD 2,000–4,000 per box	Distributed edge nodes per building/floor (≈ 30 cameras each)
Dedicated NPU accelerator cards (PCIe / E1.S)	≈ 26 TOPS per chip at ≈ 2.5 W, 4–16 chips per card or server	> 100 channels per 2U server (VMS-vendor validation)	USD 1,000–3,000 per card	Low-power alternative; models must be compiled for the NPU toolchain
Data-centre media GPUs (non-NVIDIA)	Media-engine-centric cards, ≈ 36 × 1080p60 transcode per card, OpenVINO-style inference	Decode-heavy workloads; inference below L4 class	USD 1,500–2,500	Recording-server decode offload

6.2 Purpose-built recording servers and appliances

Category	Description	Typical capacity	Notes
Storage-dense 2U rack servers (24–26 × 3.5-inch bays)	Dual server CPUs, dual 10 GbE, hardware RAID, validated by the major VMS vendors	120–300 cameras recording per server depending on bitrate and disks	Validated at 120 cameras @ 4 Mbps with 4–7 % CPU
Surveillance-specific server lines	Servers pre-tuned for VMS workloads, multi-GPU options, 5-year warranties	Up to ≈ 1,000 cameras recording-only with multi-GPU decode	Integrator standard for large sites
Turn-key VMS recording appliances	Recording servers with the VMS pre-installed and licensed	Several hundred cameras per unit	Fastest route if adopting a commercial VMS for the recording tier
AI NVR appliances (32-channel class)	Embedded deep-learning engines, ≈ 320 Mbps in, 8 disk bays, ≤ 50 W	32 cameras; face comparison 32 ch, perimeter 24 ch @ 2 MP	Appliance route: ≈ 16 units for 500 cameras; analytics tied to the vendor
Enterprise disk arrays / NAS	12 Gb SAS or 25 GbE iSCSI, 0.5–2 PB	Archive tier	Sized by retention policy

6.3 Edge-AI cameras (analytics on the camera)

Category	Capability	Server-side benefit
Cameras with a deep-learning processing unit (DLPU) on the SoC	On-camera object detection and classification, third-party app platform, AV1/H.265 with content-adaptive bitrate	Server receives metadata plus low bitrate; GPU tier only for second-stage analytics
Cameras with human/vehicle classification and perimeter analytics built in	False-alarm filtering, perimeter protection, face capture on camera	Reduces server inference by an order of magnitude
Domestic-vendor AI cameras with local support	Comparable on-camera AI with local service and language support	Simplifies procurement and maintenance

Category	Capability	Server-side benefit
ONVIF Profile M (standard)	Vendor-neutral metadata and analytics event interface	Lets “Company A” software consume edge metadata from any compliant camera

6.4 Commercial software platforms worth evaluating

If “Company A” prefers to keep its own AI models but not rebuild recording, failover, and client functions, a commercial VMS can serve as the recording tier while the in-house AI runs on the GPU tier via the VMS API:

- **Open-platform enterprise VMS** — recording servers, failover, and an analytics gateway / SDK for integrating custom analytics.
- **White-label media-server VMS** — Linux-capable media server, 256 cameras per server, open metadata SDK for third-party analytics.
- **Enterprise unified security platform** — strict per-recorder limits and strong cyber-security posture; higher licence cost.
- **NVIDIA Metropolis / DeepStream + Triton** — the analytics framework for the GPU tier regardless of VMS choice; free SDK, commercial support via NVIDIA partners.

7. Implementation Roadmap, Budget, and Risks

Phase	Timeframe	Actions	Expected result
0 – Immediate stabilisation	Weeks 1–4	Switch analytics to camera sub-streams; reduce inference to 5–10 fps; enable hardware decode; cap cameras on the current PC to a level it sustains (typically 60–100); move recording to a separate machine or NAS temporarily; add watchdog restarts and monitoring	Freezes stop; measured per-stream cost becomes known
1 – Separate recording and management	Months 1–3	Deploy 2–3 recording servers plus management/DB server and dedicated VLANs / 10 GbE; migrate cameras in batches of 50	Recording is reliable and independent of AI load
2 – GPU analytics cluster	Months 2–6	Deploy 2–3 GPU servers with NVIDIA L4 cards; port pipeline to DeepStream/TensorRT with batching; introduce MQTT/Kafka event bus and metadata database; load-test to 500 streams	AI at full camera count with N+1 redundancy
3 – Storage and archive	Months 3–6	Install archive array; implement two-tier retention; validate playback performance under load	30-day retention met at planned cost
4 – Edge-AI adoption	Months 6–12+	Specify on-board-AI cameras for new installs and replacements; integrate ONVIF Profile M metadata; shrink GPU-tier load per camera	Lower per-camera cost, headroom for more sites

7.1 Indicative budget envelope (500 cameras, hardware only)

Figures are rough order-of-magnitude estimates for budgeting and must be confirmed by quotation. Software licences (VMS, OS, database), installation, and cabling are excluded.

Item	Indicative cost (USD)	Comment
4 × recording servers with disks	80,000 – 140,000	Disk capacity dominates
3 × GPU servers with 4 × NVIDIA L4 each	60,000 – 90,000	12 × L4 ≈ 35,000–40,000 of this
2 × management/DB servers	15,000 – 25,000	
Archive storage ≈ 650 TB usable	60,000 – 120,000	Halve with event-based recording
Network (10/25 GbE core, PoE access)	20,000 – 50,000	Depends on existing switches
UPS, rack, cabling	10,000 – 20,000	
Total hardware	≈ 245,000 – 445,000	≈ USD 500–900 per camera

7.2 Risks and mitigations

Risk	Mitigation
AI models not portable to TensorRT / batched pipeline	Start with ONNX export; use Triton for models that cannot be converted; keep one model per GPU worker as fallback
Network already saturated or shared with office traffic	Dedicated VLANs and 10 GbE uplinks are prerequisites of Phase 1
Camera sub-streams not configured or too low quality	Audit all cameras; ONVIF bulk configuration tools; set 720p/10 fps as standard
Single-site power/cooling for a 6–8 kW rack	Confirm with facilities; NVIDIA L4 (72 W) chosen to keep power low
Team unfamiliar with Linux/containers/Kubernetes	Begin with Docker Compose on each GPU server; adopt Kubernetes only when > 3 nodes
Vendor lock-in with AI NVR appliances	Prefer an open GPU tier plus ONVIF Profile M cameras; use appliances only for isolated small sites

8. Conclusion

The observed freezes are the predictable outcome of running an ingest, recording, and multi-model inference workload for several hundred cameras on one machine. Every major surveillance product solves this the same way: dedicated recording servers limited to a few hundred cameras each, a separate GPU or NPU analytics tier sized at tens of streams per accelerator, an isolated management/database layer, tiered storage, and a dedicated 10 GbE network — increasingly complemented by cameras that perform first-pass AI themselves. Adopting the same tiered architecture, with NVIDIA L4-based DeepStream analytics servers and purpose-built recording

servers, will give a platform that reliably supports 500 cameras today and scales to further sites by adding nodes rather than replacing them.

The recommended next steps are to (1) apply the Phase 0 stabilisation changes and measure the real per-stream decode and inference cost of the current models, (2) request quotations for the reference bill of materials in Section 5.3 from server and NVIDIA distribution partners, and (3) run a 50-camera pilot of the DeepStream pipeline on a single L4 server to validate the sizing assumptions before committing to the full cluster.

Note on sources: capacity figures in Section 4 are taken from publicly published sizing guides, white papers, and validation reports of the leading VMS, camera, and server vendors and from NVIDIA's DeepStream and Video Codec SDK documentation. Vendor names have been omitted from this document by design; the full reference list is available from the project team on request. Indicative prices and effective streams-per-GPU planning figures are engineering estimates, not vendor quotations.